

Cirque Corporation

Network Security Management Procedure

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1. Purpose

The purpose of this procedure is to define the systematic process for managing the security of Cirque's networks, including wired, wireless, and remote access connections. This procedure aims to protect network services and information from unauthorized access, misuse, disclosure, or disruption, in accordance with IS-CIRQ-P-011-G: Operations Security Policy and ISO/IEC 27001:2022 Annex A.8.19.

****2. Scope****

2. Scope

This procedure applies to all Cirque network infrastructure, devices, and services across all locations, including the US, Japan, and China offices, and remote access points. This includes, but is not limited to:

- Network devices (routers, switches, firewalls, wireless access points).
- Network services (DNS, DHCP).

- Network protocols.
- Network cabling.
- Virtual Private Network (VPN) solutions.
- Cloud network configurations (e.g., Azure VNets).

****3. Responsibilities****

3. Responsibilities

- ****IT Manager:**** Overall owner of this procedure. Responsible for network architecture, security design, approval of network changes, and oversight of network security monitoring.
- ****System Administrators / IT Personnel:**** Responsible for implementing, configuring, maintaining, and monitoring network devices and services in accordance with this procedure.
- ****All Personnel:**** Responsible for adhering to network usage policies, especially concerning remote access.

****4. Procedure****

4. Procedure

****4.1. Network Segmentation and Architecture**** a. Cirque networks shall be logically segmented to isolate different functional areas, sensitive data, and system classifications (e.g., corporate network, production network, development network, guest wireless). b. Network segmentation shall be enforced using firewalls and VLANs (Virtual Local Area Networks). c. Network diagrams, illustrating segmentation and key security zones, shall be maintained and regularly updated.

****4.2. Firewall Management**** a. ****Firewall Deployment:**** Firewalls shall

be deployed at critical network boundaries, including the perimeter between Cirque's internal network and external networks (e.g., the Internet) and between network segments. b. **Rule Management:** Firewall rules shall be explicitly defined, approved by the IT Manager, and based on a least-privilege principle (deny-all by default, permit-by-exception). Rules shall be documented, including justification, source/destination IP, port, and protocol. Rules shall be regularly reviewed (at least quarterly) and unnecessary rules removed. c. **Secure Configuration:** Firewalls shall be securely configured, with default passwords changed, unnecessary services disabled, and management interfaces protected. d. **Logging:** Firewall logs shall be enabled and forwarded to a centralized logging system as per IS-CIRQ-PR-015-G: Logging and Monitoring Procedure.

4.3. Secure Network Device Configuration a. **Hardening:** All network devices (routers, switches, wireless access points) shall be hardened according to industry best practices and vendor recommendations. This includes: Changing all default passwords. Disabling unnecessary services and ports. Implementing strong authentication and authorization for device management. Restricting management access to designated administrative networks or hosts. b. **Patch Management:** Network device firmware and operating systems shall be regularly updated with security patches. c. **Configuration Backup:** Device configurations shall be regularly backed up and stored securely.

4.4. Wireless Network Security a. **Strong Encryption:** Wireless networks used for corporate access shall utilize strong encryption protocols (e.g., WPA2 Enterprise or WPA3) with EAP-TLS or PEAP for user authentication. b. **Separate Guest Network:** A separate, isolated guest wireless network shall be provided for visitors, ensuring it has

no access to Cirque\’s internal corporate or production networks. c.

****Authentication:**** All wireless access points shall require strong authentication.

****4.5. Remote Access Security (VPN)**** a. ****Mandatory VPN:**** All remote access to Cirque\’s internal networks and resources shall be conducted exclusively through an approved Virtual Private Network (VPN) solution.

b. ****Strong Encryption:**** The VPN solution shall use strong cryptographic protocols (e.g., IPsec, OpenVPN, or TLS VPN) for data in transit encryption as per IS-CIRQ-P-009-G: Cryptography Policy. c.

****Multi-Factor Authentication (MFA):**** MFA shall be mandatory for all VPN connections. d. ****Device Compliance:**** Only company-managed devices (e.g., laptops managed by ****Intune****) with current security software (****Windows Defender for Business****) and up-to-date patches shall be permitted to connect via VPN.

****4.6. Network Monitoring**** a. ****Traffic Analysis:**** Network traffic shall be monitored for suspicious activity, intrusions, and anomalies using network intrusion detection/prevention systems (IDS/IPS) or similar tools. b. ****Bandwidth Monitoring:**** Network bandwidth usage shall be monitored to ensure adequate capacity and detect unusual traffic patterns. c. ****Alerting:**** Alerts shall be configured for critical network security events and performance issues.

****4.7. Network Service Security**** a. ****DNS Security:**** DNS services shall be configured securely, and internal DNS servers protected from external access. b. ****DHCP Security:**** DHCP services shall be managed to prevent unauthorized IP address assignment. c. ****Time Synchronization:**** Network devices shall be synchronized to a central, reliable time source (NTP) to ensure accurate logging.

****4.8. Segregation of Duties for Network Administration**** a. Where

feasible, duties related to network administration (e.g., firewall rule changes, router configuration) shall be segregated to prevent a single individual from being able to compromise network security without detection.

****5. Review and Update****

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This procedure will be reviewed at least annually, or sooner if there are significant changes to Cirquel's network architecture, security technologies, or the threat landscape.

****6. Related Documents****

6. Related Documents

- IS-CIRQ-P-011-G: Operations Security Policy
- IS-CIRQ-P-008-G: Access Control Policy
- IS-CIRQ-PR-008-G: Access Control Procedure
- IS-CIRQ-P-009-G: Cryptography Policy
- IS-CIRQ-PR-015-G: Logging and Monitoring Procedure
- IS-CIRQ-PR-013-G: Change Management Procedure